Chapter 6: Life Processes

1. Introduction

Living beings are distinguished from non-living things by various observable movements (like walking, breathing, growing). However, underlying all these are invisible molecular movements. **Life Processes** are the basic, fundamental functions performed by living organisms to maintain their life on Earth.

The primary life processes are **Nutrition, Respiration, Transportation, and Excretion**.

2. Nutrition

Nutrition is the process of taking in food and utilizing it to produce energy required for performing various tasks.

Types of Nutrition:

- **Autotrophic Nutrition:** Organisms (like green plants and some bacteria) make their own food from simple inorganic substances (CO2 and H2O) using sunlight.
- **Heterotrophic Nutrition:** Organisms depend on other organisms for their food (e.g., animals, fungi).

Nutrition in Plants (Photosynthesis)

> **Photosynthesis formula:**
> $6CO_2 + 12H_2O + (\text{Light Energy/Chlorophyll}) \rightarrow C_6H_{12}O_6 \text{ (Glucose)} + 6O_2 + 6H_2O$

Events during Photosynthesis:

1. Absorption of light energy by chlorophyll.

2. Conversion of light energy to chemical energy and splitting of water molecules into hydrogen and oxygen.
3. Reduction of carbon dioxide to carbohydrates.

(Diagram Explanation: A leaf cross-section shows the upper and lower epidermis. Guard cells flank the stomata on the lower surface, which controls gas exchange and transpiration. Chloroplasts contain chlorophyll where photosynthesis occurs.)

Nutrition in Human Beings

The human digestive system consists of the alimentary canal and associated glands.

- **Mouth:** Digestion starts here. Salivary amylase breaks starch into sugar.
- **Oesophagus:** Pushes food down to the stomach via peristaltic movements.
- **Stomach:** Gastric glands secrete HCl, Pepsin (digests protein), and Mucus.
- **Small Intestine:** Site of complete digestion. Receives bile from the **Liver** (emulsifies fats) and pancreatic juice from the **Pancreas**. Inside the small intestine, finger-like projections called **villi** increase the surface area for absorption.
- **Large Intestine:** Absorbs excess water.

3. Respiration

The process of acquiring oxygen from outside the body, and using it in the process of break-down of food sources for cellular needs, is called respiration.

Breakdown of Glucose by Various Pathways:

First step: Glucose (6C) → Pyruvate (3C) in the cytoplasm.
Subsequent breakdown depends on the presence of oxygen:

- **Aerobic Respiration:** Complete breakdown of pyruvate in the presence of oxygen (in mitochondria) generating 36/38 ATP, CO_2, and water.
- **Anaerobic Respiration (in yeast):** Absence of oxygen generates Ethanol, CO_2, and limited energy (2 ATP).
- **Lack of Oxygen (in human muscle cells):** Generates Lactic acid and energy, causing muscle cramps.

4. Transportation

Transportation in Humans

Managed by the circulatory system: Heart, Blood, and Blood Vessels.

- **The Heart:** A muscular organ with 4 chambers. The right atrium receives deoxygenated blood, pumps it to the right ventricle, which sends it to the lungs. Oxygenated blood from lungs enters the left atrium, moves to the left ventricle, which pumps it to the rest of the body.
- **Blood Vessels:** *Arteries* carry blood away from the heart (thick walls). *Veins* carry blood towards the heart (contain valves to prevent backflow). *Capillaries* are where exchange of materials takes place.

Transportation in Plants

- **Xylem:** Transports water and minerals from roots to other parts. Works via transpiration pull.
- **Phloem:** Transports food (amino acids and sugars) from leaves to other parts. This process is called **translocation**.

5. Excretion

The biological process involved in the removal of harmful metabolic wastes from the body is called excretion.

- **Kidneys:** The main excretory organs in humans. They filter wastes (like urea) from the blood. The basic filtration unit is the **nephron**.
- **Excretion in Plants:** Oxygen, CO2, and water are released via stomata. Other wastes are stored in leaves (which fall off) or as resins and gums.

6. Key Points / Important Notes

- **Villi** are richly supplied with blood vessels which take the absorbed food to each and every cell of the body.
- **Alveoli** in the lungs provide a massive surface area where the exchange of gases easily takes place.
- Plants use low energy and slow transport systems because they do not move and have large proportions of dead cells.

7. Frequently Asked Questions

Q. Why do herbivores have a longer small intestine than carnivores?

Ans. Herbivores eat grass, which contains cellulose. Cellulose takes longer to digest, thus requiring a longer intestine.

Q. What is the role of acid in our stomach?

Ans. Hydrochloric acid (HCl) kills harmful bacteria entering with food and creates an acidic medium which facilitates the action of the enzyme pepsin.

8. Multiple Choice Questions (MCQs)

1. Which of the following is an autotroph?

a. Green plant

b. Fungus

c. Human

d. Amoeba

Answer: a) Green plant

2. The site of photosynthesis in the cells of a leaf is:

a. Chloroplast

b. Mitochondria

c. Cytoplasm

d. Protoplasm

Answer: a) Chloroplast

3. In amoeba, food is digested in the:

a. Food vacuole

b. Mitochondria

c. Pseudopodia

d. Chloroplast

Answer: a) Food vacuole

4. Which of the following events in the mouth cavity will be affected if salivary amylase is lacking in the saliva?

a. Proteins breaking down into amino acids

b. Starch breaking down into sugars

c. Fats breaking down into fatty acids and glycerol

d. Absorption of vitamins

Answer: b) Starch breaking down into sugars

5. Which region of the alimentary canal absorbs the digested food?

a. Stomach

b. Small intestine

c. Large intestine

d. Liver

Answer: b) Small intestine

6. Complete digestion of carbohydrates, proteins and fats takes place in:

a. Stomach

b. Liver

c. Small intestine

d. Large intestine

Answer: c) Small intestine

7. The breakdown of pyruvate to give carbon dioxide, water and energy takes place in:

a. Cytoplasm

b. Mitochondria

c. Chloroplast

d. Nucleus

Answer: b) Mitochondria

8. Glycolysis process occurs in which part of the cell?

a. Cytoplasm

b. Nucleus

c. Mitochondria

d. Chloroplast

Answer: a) Cytoplasm

9. Name the substances whose build-up in the muscles during vigorous physical exercise may cause cramps?

a. Ethanol + Carbon dioxide + Energy

b. Lactic acid + Energy

c. Carbon dioxide + Water + Energy

d. Pyruvate

Answer: b) Lactic acid + Energy

10. Name the pores in a leaf through which respiratory exchange of gases takes place.

a. Lenticels

b. Vacuoles

c. Xylem

d. Stomata

Answer: d) Stomata

11. The respiratory pigment in human beings is:

a. Carotene

b. Chlorophyll

c. Haemoglobin

d. Mitochondria

Answer: c) Haemoglobin

12. The movement of food in phloem is called:

a. Transpiration

b. Translocation

c. Respiration

d. Evaporation

Answer: b) Translocation

13. Which plant tissue transports water and minerals from the roots to the leaf?

a. Xylem

b. Phloem

c. Parenchyma

d. Collenchyma

Answer: a) Xylem

14. Blood consists of what fluid medium?

a. Lymph

b. Platelets

c. Plasma

d. All of these

Answer: c) Plasma

15. Which part of blood carries dissolved molecules of food, CO_2, and nitrogenous wastes?

a. RBC

b. WBC

c. Plasma

d. Platelets

Answer: c) Plasma

16. The correct path of urine is:

a. Kidney -> Ureter -> Urethra -> Urinary bladder

b. Kidney -> Urinary bladder -> Urethra -> Ureter

c. Kidney -> Ureter -> Urinary bladder -> Urethra

d. Urinary bladder -> Kidney -> Ureter -> Urethra

Answer: c) Kidney -> Ureter -> Urinary bladder -> Urethra

17. Single circulation, i.e., blood flows through the heart only once during one cycle of passage through the body, is exhibited by:

a. Labeo, Chameleon, Salamander

b. Hippocampus, Exocoetus, Anabas (Fishes)

c. Hyla, Rana, Draco

d. Whale, Dolphin, Turtle

Answer: b) Hippocampus, Exocoetus, Anabas (Fishes)

18. Name the tube which connects the kidneys to the urinary bladder.

a. Urethra

b. Nephron

c. Tubule

d. Ureter

Answer: d) Ureter

19. Which part of nephron allows the selective reabsorption of useful substances like glucose, amino acids, salts and water into the blood capillaries?

a. Tubule

b. Glomerulus

c. Bowman's capsule

d. Ureter

Answer: a) Tubule

20. In plants, the waste products are stored as:

a. Resins

b. Gums

c. Essential oils

d. All of the above

Answer: d) All of the above

9. Short Answer Questions

1. State two functions of stomata.

Answer: 1) They allow the exchange of gases (CO_2 and O_2) with the atmosphere. 2) They facilitate transpiration (loss of water vapor), which helps in the upward pull of water in xylem.

2. How is small intestine designed to absorb digested food?

Answer: The inner lining of the small intestine has numerous finger-like projections called villi. These villi greatly increase the surface area for absorption and are richly supplied with blood vessels that carry absorbed food to all parts of the body.

3. What is the advantage of a four-chambered heart in humans?

Answer: A four-chambered heart completely separates oxygenated and deoxygenated blood. This highly efficient supply of oxygen is useful in animals that have high energy needs, such as birds and mammals, which constantly use energy to maintain body temperature.

10. Long Answer Questions

1. Describe the structure and functioning of nephrons.

Answer: Nephrons are the functional and structural filtering units of the kidneys.

Structure: Each nephron consists of a cup-shaped structure called Bowman's capsule containing a bundle of capillaries known as the glomerulus. The Bowman's capsule leads into a long tubular structure which eventually joins a collecting duct.

Functioning:
1. *Filtration:* Blood under high pressure flows through the glomerulus. Water, urea, glucose, and salts are filtered into Bowman's capsule.
2. *Reabsorption:* As the filtrate passes through the tubular part, useful substances like glucose, amino acids, and major amounts of water are selectively reabsorbed by surrounding blood capillaries.
3. *Secretion:* Certain substances like potassium ions are secreted into the tubule. The final liquid entering the collecting duct is urine.

2. Explain the process of digestion in the human small and large intestine.

Answer:

Small Intestine: Chyme (partially digested food) enters the small intestine. Liver secretes bile which emulsifies large fat globules into smaller ones. The pancreas secretes pancreatic juice containing trypsin for digesting proteins and lipase for breaking down emulsified fats. Intestinal juice enzymes finally convert complex carbohydrates into glucose, proteins into amino acids, and fats into fatty acids and glycerol. Villi then absorb these simple nutrients.

Large Intestine: The unabsorbed and undigested food is sent to the large intestine. The walls of the large intestine absorb excess water from this material. The remaining solid waste is stored in the rectum and expelled out via the anus.